

Livestream Operational Intelligence Integration

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Objective

Transform livestreaming from a Socket.IO-only feature into an Operational Intelligence data source for YenkaSAI.

Required architecture:

Livestream Client -> Socket Layer -> Livestream Event Service -> Operational Events Layer (OIL) -> YenkaSAI

Findings

1. Livestream Socket.IO sync already existed, but events were emitted as older lowercase YME events such as 'live_comment', 'live_reaction', 'live_joined', and 'gift_sent'.
2. The current real-time path preserved UX but did not consistently expose canonical platform event types such as 'STREAM_COMMENT', 'STREAM_GIFT', 'STREAM_REPORT', or 'STREAM_VIEW_DURATION'.
3. HTTP livestream routes handled important actions outside the socket path, especially join-token requests, host end, auto-end, viewer leave, and gifts.
4. The AI event relay was configured for '/api/events', while the requested OIL ingestion target is '/api/events/ingest'.
5. YME scoring, consolidation, interest extraction, and recommendation signals did not know the new stream event names.

Fixes Implemented

1. Canonical Livestream Operational Event Service

Created 'src/services/livestream/operationalEvents.service.js'.

This service defines all required event types:

- * 'STREAM_STARTED'
- * 'STREAM_ENDED'
- * 'STREAM_JOINED'
- * 'STREAM_LEFT'
- * 'STREAM_COMMENT'
- * 'STREAM_LIKE'
- * 'STREAM_GIFT'
- * 'STREAM_SHARE'

- * 'STREAM_REPORT'
- * 'STREAM_FOLLOW_HOST'
- * 'STREAM_PIN_COMMENT'
- * 'STREAM_MODERATION_ACTION'
- * 'STREAM_BAN_USER'
- * 'STREAM_WARNING'
- * 'STREAM_VIEW_DURATION'
- * 'STREAM_PEAK_VIEWERS'

It emits the canonical schema:

```
{  
  "eventType": "STREAM_COMMENT",  
  "streamId": "...",  
  "userId": "...",  
  "hostId": "...",  
  "timestamp": "...",  
  "metadata": {}  
}
```

Evidence:

- * Event type constants: 'src/services/livestream/operationalEvents.service.js:4'
- * Schema builder: 'src/services/livestream/operationalEvents.service.js:163'
- * OIL/YME publishing path: 'src/services/livestream/operationalEvents.service.js:284'

2. Analytics Metrics

The operational event service now populates livestream metrics:

- * 'totalViewers'
- * 'concurrentViewers'
- * 'peakViewers'
- * 'commentsPerMinute'
- * 'likesPerMinute'
- * 'giftsPerMinute'
- * 'watchTime'
- * 'averageWatchTime'
- * 'retentionRate'
- * 'giftRevenue'
- * 'topGifters'
- * 'topStreams'

Evidence:

- * Metrics state and public metrics:
'src/services/livestream/operationalEvents.service.js:46'
- * Metrics update logic: 'src/services/livestream/operationalEvents.service.js:98'
- * Top streams accessor: 'src/services/livestream/operationalEvents.service.js:335'
- * API metrics endpoints: 'routes/livestream.routes.js:449'

3. Moderation Processing

Livestream events now generate moderation signals for:

- * spam comments
- * gift fraud
- * bot viewers
- * abuse reports
- * suspicious engagement spikes

Detected signals create 'ModerationItem' records with 'source: livestream_operational_intelligence'.

Evidence:

- * Moderation signal detection: 'src/services/livestream/operationalEvents.service.js:185'
- * Moderation queue item creation: 'src/services/livestream/operationalEvents.service.js:244'

4. Socket.IO Room Preservation and New Broadcasts

Socket.IO rooms now include the required room shape:

- * 'stream:{streamId}'

Compatibility rooms remain active:

- * 'livestream_{streamId}'
- * 'live:{streamId}'

Real-time broadcasts now include the requested names:

- * 'new_comment'
- * 'new_like'
- * 'new_gift'
- * 'viewer_joined'
- * 'viewer_left'
- * 'gift_animation'

Evidence:

- * Room update: 'src/services/livestream/livestream.service.js'
- * 'new_comment': 'src/services/livestream/livestream.socket.js:412'
- * 'new_like': 'src/services/livestream/livestream.socket.js:437'
- * 'viewer_joined': 'src/services/livestream/livestream.socket.js:313'
- * 'viewer_left': 'src/services/livestream/livestream.socket.js:356'
- * 'new_gift' and 'gift_animation': 'routes/livestream.routes.js:757'

5. Socket Layer Event Coverage

Socket handlers now emit operational events for:

- * start
- * join
- * leave
- * comment
- * like
- * share
- * report
- * follow host
- * pin comment
- * moderation action
- * ban user
- * warning
- * view duration
- * disconnect watch duration

Evidence:

- * Shared socket operational emitter: 'src/services/livestream/livestream.socket.js:10'
- * Comment and like mapping: 'src/services/livestream/livestream.socket.js:394'
- * Share, report, follow-host, pin-comment, moderation actions: 'src/services/livestream/livestream.socket.js:445'
- * Disconnect leave and view duration capture: 'src/services/livestream/livestream.socket.js:853'

6. HTTP Route Event Coverage

HTTP routes now emit operational events for:

- * livestream join token
- * host end
- * auto-end
- * startup failure
- * leave
- * watch duration
- * gift

Evidence:

- * HTTP operational emitter: 'routes/livestream.routes.js:171'
- * Join token event: 'routes/livestream.routes.js:558'
- * End event: 'routes/livestream.routes.js:621'
- * Leave and view-duration event: 'routes/livestream.routes.js:659'
- * Gift event: 'routes/livestream.routes.js:773'

7. OIL and YenkaSAI Relay

The intelligence relay default ingest path now targets:

`'/api/events/ingest'`

The relay supports all canonical livestream events as normalized 'stream_*' operational events and forwards livestream metadata including stream ID, host ID, metrics, moderation signals, and recommendation signals.

Evidence:

- * Default ingest path: 'src/intelligence/services/eventPublisher.service.js:8'
- * Supported stream event types: 'src/intelligence/services/eventPublisher.service.js:54'
- * Livestream relay metadata mapping:
'src/intelligence/services/eventPublisher.service.js:576'

8. YME Memory and Recommendation Integration

YME now recognizes stream event types in:

- * source app config
- * event ingestion aliases
- * importance scoring
- * interest extraction
- * recommendation scoring
- * consolidation trigger list

Evidence:

- * Source apps and event types: 'src/yme/config/yme.config.js'
- * Event aliases: 'src/yme/services/eventIngestion.service.js'
- * Importance scoring: 'src/yme/services/importanceScoring.service.js'
- * Interest weights: 'src/yme/services/interestExtraction.service.js'
- * Recommendation weights: 'src/yme/services/recommendationSignals.service.js'
- * Consolidation triggers: 'src/yme/services/consolidation.service.js'

Verification Evidence

Command:

```
npm test
```

Result:

```
tests 10  
pass 10  
fail 0  
duration_ms 10022.635969
```

Additional syntax checks passed:

```
node --check src/services/livestream/livestream.socket.js  
node --check routes/livestream.routes.js  
node --check src/services/livestream/operationalEvents.service.js
```

```
node --check src/intelligence/services/eventPublisher.service.js
```

Focused test evidence:

- * Canonical livestream schema test: 'tests/livestreamOperationalEvents.test.js:11'
- * Metrics population test: 'tests/livestreamOperationalEvents.test.js:33'
- * AI relay 'STREAM_GIFT' mapping test: 'tests/intelligenceEventPublisher.test.js'

Implementation Notes

The new metrics store is in-process. It is suitable for immediate operational intelligence and creator dashboard snapshots, but production-grade historical analytics should later persist rollups into MongoDB or a streaming analytics store.

The implementation preserves current client compatibility by keeping existing 'live_*' broadcasts while adding the requested event names.

Changed Files

- * 'src/services/livestream/operationalEvents.service.js'
- * 'src/services/livestream/livestream.socket.js'
- * 'src/services/livestream/livestream.service.js'
- * 'routes/livestream.routes.js'
- * 'src/intelligence/services/eventPublisher.service.js'
- * 'src/yme/config/yme.config.js'
- * 'src/yme/services/eventIngestion.service.js'
- * 'src/yme/services/importanceScoring.service.js'
- * 'src/yme/services/interestExtraction.service.js'
- * 'src/yme/services/recommendationSignals.service.js'
- * 'src/yme/services/consolidation.service.js'
- * 'tests/livestreamOperationalEvents.test.js'
- * 'tests/intelligenceEventPublisher.test.js'

Conclusion

Livestreaming is now wired as an Operational Intelligence source. Real-time synchronization remains intact, and every major livestream action can now feed analytics, moderation, recommendations, YME memory, security detection, and the YenkaSAI event relay.